ODP 5: Functional Prototype LS1&2

Design Documentation:
Parts List:

Part Name	McMaster Code/Amazon Link	Specs	Fabrication
Bearing	6384K365	Flanged ball, ⅝" shaft dia, 1 ⅜" housing dia, steel	N/A
Switch	7343K184-7343K186	28V DC, Maintained, 2 Terminals (On/Off), Screw Terminals	N/A
Solar Panel (3x)	Solar Panels	4.3"x2.4" cells, 11.8" cable, DC 6V 1W.	N/A
Solar Power Adapter	Power Adapter	Type C, Solar - MCU	N/A
Boost Converter	6V Boost Converter	2V-24V to 5V-28V, pack of 10 (1 needed)	N/A
Speed Controller	Speed Controller	1.8-12V, DC	N/A
Motor	Motor	3mm shaft, 6V DC 100 RPM	N/A
Rotor	N/A - Self-Fabricated	240mm dia, 30mm height, 3 blades, patterned extrusions on underside	3D Printed
Top Drum	N/A - Self-Fabricated	254 mm dia, 60mm height. Centered hole 60mm dia, circular patterned extrusions	3D Printed
Bottom Drum	N/A - Self-Fabricated	254mm dia, 125 mm height	3D Printed
Electronics Bay	N/A - Self-Fabricated	254mm dia, 45 mm height	3D Printed
Shaft	N/A - Self-Fabricated	Couples to 3mm motor and 15mm rotor hex, with 10	3D Printed

		mm main shaft (195mm total length)	

Design Intent:

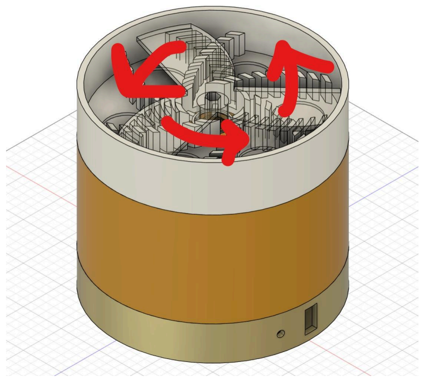


Figure 1: Full assembly and motion of 3D printed pieces

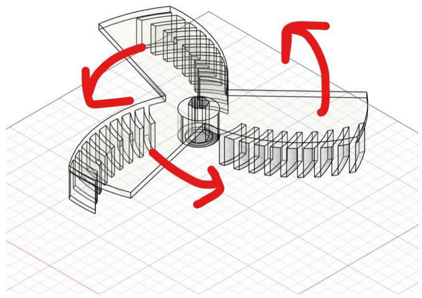


Figure 2: Motion and design of 3D printed spinner piece

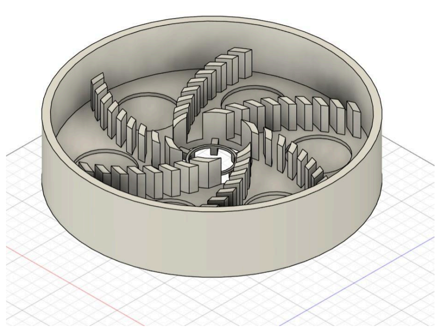


Figure 3: Design of 3D printed top drum piece

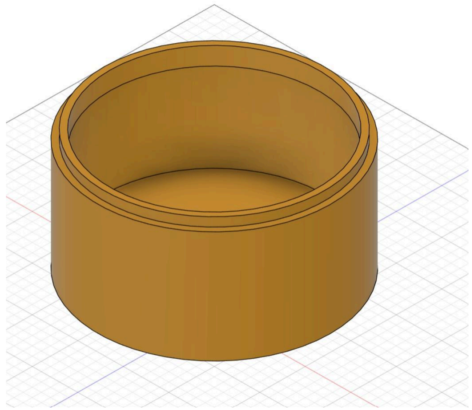


Figure 4: Design of 3D printed bottom drum piece

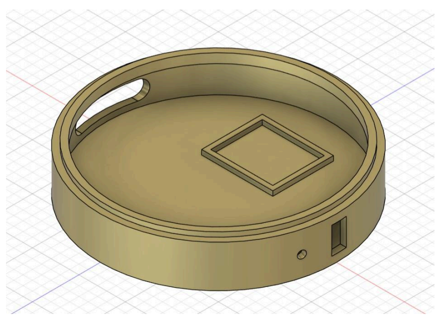
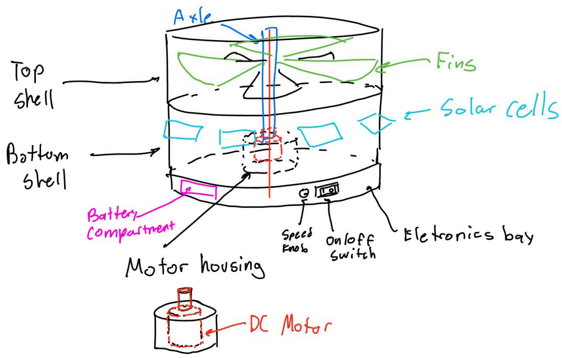
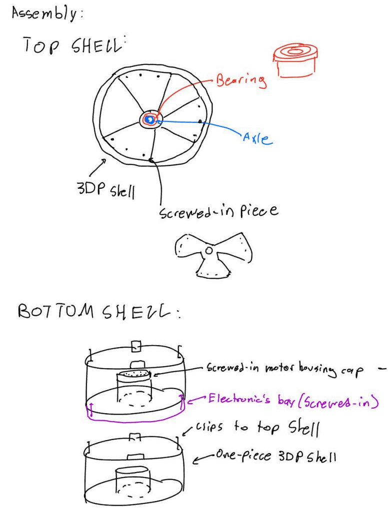


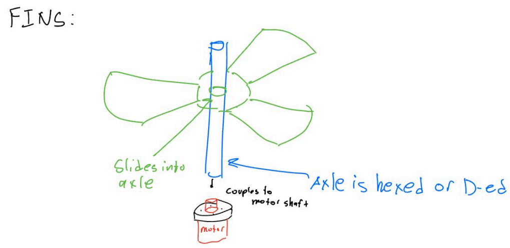
Figure 5: Design of 3D printed electronics bay

Design Assembly Instructions:

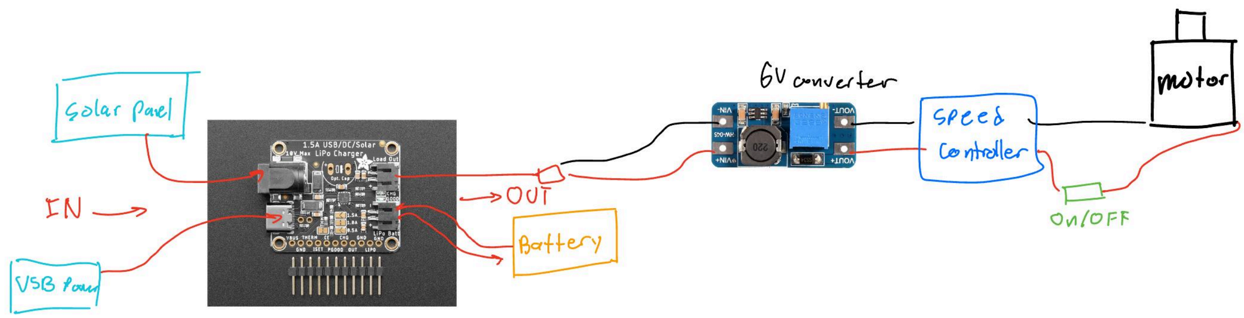
Bottom-up assembly: Electronics bay -> Bottom Drum -> Top Drum -> Rotor. Internally: All electronics except for the motor will be stored in compartments within the electronics bay. The bay, bottom drum, and top drum will "stack" onto each other using gravity to constrain z translation, and an internal constraint extrusion to limit x and y translation. The motor screws into the bottom drum's motor housing compartment, with the axle sticking upwards. This will be connected to the main axle (press fit), which will connect to the rotor (hex joint, with a press fit or permanently melted together). The upper connection passes through the hole in the center of the top drum, such that the rotor is vertically supported by the inner ring of the ball bearing fixed to the drum. The motor drives the axle, which simply drives the rotor.







Electronic assembly:



The motor, speed controller, and switch are connected. The speed controller connects to the converter, both of which are connected to the LiPo charger (which is connected to the battery, solar panels, and optional USB Power for testing). All of these components “live” in the electronics bay freely, despite the motor being fixed to the bottom drum. There is ample open space for the connections to be made (including to the motor).

Design Testing:

Test 1: Motor & Power source

- This test is focused on the motor and an arbitrary power source to check if the battery is able to drive the motor and the motor is able to drive the rotation of the system (the rotor).
- This test is accomplished by assembling our system (connecting the motor to a power source and switch, connecting it to an axle and secondarily to the rotor, all

constrained by the container components of the trap) and flipping the switch, then checking to see if the system rotates as expected.

- The test results were conclusive and positive - we were able to drive the motor at ~100 rpm using a 6V power supply, which turned the motor shaft alone. When coupled with the extended shaft and the rotor, the motor supplied enough torque to drive the rotor at the desired speed (100 rpm). Wear and tear have been accounted for by minimizing friction and operating at low speeds (using a ball bearing and a slow rpm DC motor). Furthermore, we confirmed that the motor can operate continuously - the expected function in the field.
- For the next iteration, we will optimize the motor mount, axle connections, and rotor path (eliminate unnecessary friction) to ensure smooth operation. We will also test different RPMs using the speed controller.

Test 2: Solar panels

- This test is purely focused on the solar panels (the cells), checking that they are able to generate enough voltage to power our system. There are 2 stages of this set of testing: solar panel voltage output (initial test), and capacity for the panel to drive the motor (secondary test).
- The primary/initial test is accomplished by hooking up a multimeter to the solar cell-power converter system to ensure at least a 6V output, while the cell is lit by sunlight. The secondary test is accomplished by disconnecting the system from the battery source and taking it outside into the sun/well lit window area, and checking to see if the motor rotates.
- The primary test results were conclusive and positive. When the multimeter was connected to the solar cell-power converter assembly, it gave a reading of 6V, which is enough to power our 6V DC motor. The secondary test was inconclusive - we were unable to directly power the motor using solar power. However, this test showed that the issue with the solar power component was downstream from the cell itself and the power converter.
- For the next iteration, it may be worthwhile to consider a simply battery powered option if the solar power continues to be finicky. We will continue to experiment with troubleshooting the connectivity downstream from the solar cell, but will consider pivoting to a chargeable/replaceable battery pack to power the motor instead (using a 6V battery pack).

Success Criteria:

The main purpose of our device is to reduce the amount of spotted lantern flies that infest grape farms around a 1-meter radius, which would increase the rate of grape production for the user. The way the device achieves this is by luring, trapping, and killing lantern flies with a rotary mechanism that pushes the SLF into a secure enclosure, lured in by an attractant such as Tree of Heaven sap. We can quantify our success with the prototype through how effective the device is at these three tasks:

1. Luring (#of SLF attracted per unit time)
2. Trapping (% of lured SLF which are subsequently trapped)
3. Killing (% of SLF that become trapped and die as a result)

Due to the nature of this project and the resources at our disposal, it is not possible to do meaningful tests for luring and killing, since we do not have access to live lantern flies. So, for our prototype (further iterations including the presented exhibit deliverable), we will measure success through how well the device can trap mock SLF, made from material that would accurately emulate the size and weight of a real lantern fly (~3-20mm, ~1g pieces of paper representing different stages of SLF lifecycle). This will test both the motor capabilities and how well the fin geometry works at pushing lantern flies into our enclosure. These achievements can be demonstrated clearly during exhibit day.

The device should be cheap, easily manufactured/assembled, and effective.

- The device parts should be mostly 3D printed or ordered, and should not require uncommon tools to assemble.
- The device should be easy to carry: it should be less than 20 pounds and take up less than a cubic foot of space. Getting lighter/smaller is not a priority, especially if these conditions infringe on effectiveness.
- Maintenance assembly (not including putting together electronics or pieces meant to remain attached from day-to-day) should be achievable in under 2 minutes, with shorter times not being a priority.
- The device should be able to trap in the bottom drum more than 50% of SLF that have been lured in, with higher effectiveness rates being a priority.